Mechanical Dock Leveler

USER MANUAL



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1. INTRODUCTION

Please read this manual carefully before starting, operating, repairing or maintaining the Dock Levelers. The manual describes the installation, functioning, repair and maintenance of your Dock Levelers, as well as its use and the safety measures and precautions to be taken. Contact EP if you require more information.

The operator and maintenance staff should read this manual and follow training before working with this machine.

Keep this manual throughout the service life of the Dock Levelers. The operator and all persons who come into contact with the Dock Levelers should read it regularly and have access to it at all times.

Failure to comply with the safety instructions and warnings in this manual could result in serious injury to the operator or other persons in the vicinity of the machine.

Dangers to people are indicated as follows:

 **warning**

Points for consideration for the proper operation and long service life of your Dock Leveler are indicated as follows:

 **note**

People with responsibility for the safe and proper operation of this Dock Levelers must use the procedures, working practices that are described in this manual.

2. SAFETY WARNINGS

While working with the Dock Leveler, the person who is responsible for supervising the operation environment should take the necessary organizational measures to guarantee the employees' health and safety as required by local legislation.

This includes:

2.1 Appointment of authorised personnel

Authorised personnel are staff who are permitted to operate a Dock Leveler according to local statutory requirements and are appointed within the company to do this.

2.2 Definition of danger zone

The danger zone is the area within which people may be at risk from the movements of the Dock Levelers. Only the operator of the Dock Levelers may be in the danger zone.

2.3 Prevention of injury

Operating procedures and instructions should be introduced, that as far as reasonably practicable prevent the risk of injury to persons in the working area of the Dock Leveler. These include:

- never make adjustments to the machine if you are not trained and authorized to do so; avoid making adjustments to the machine when the machine is moving; never allow people near the Dock Leveler when you are carrying out work;
- provide additional illumination if the visibility in the working area is restricted;

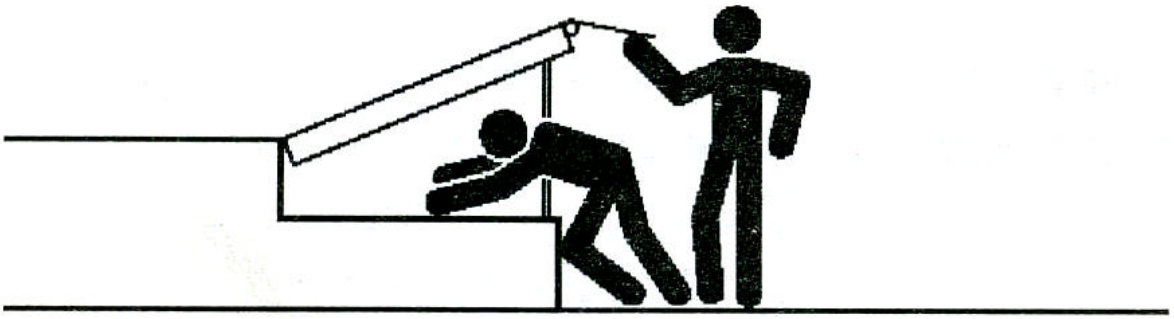
Consult your prevention adviser if desired.

2.4 Operational Satety Precautions

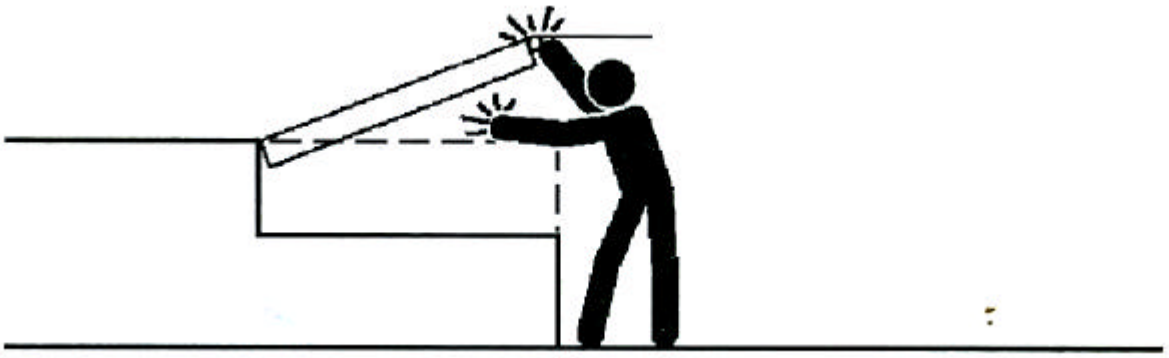
Warning



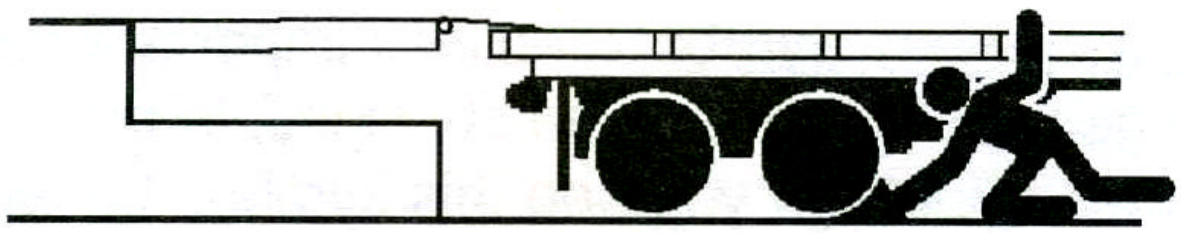
2.4.1 Stay clear of dock leveller when freight carrier is entering or leaving area.



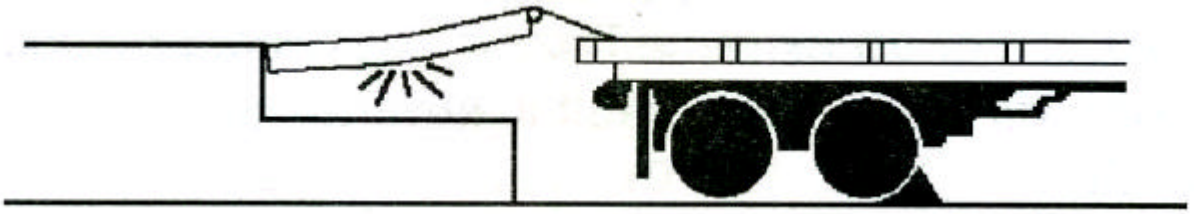
2.4.2 Do not move or use the dock leveller if anyone is under or in front of it.



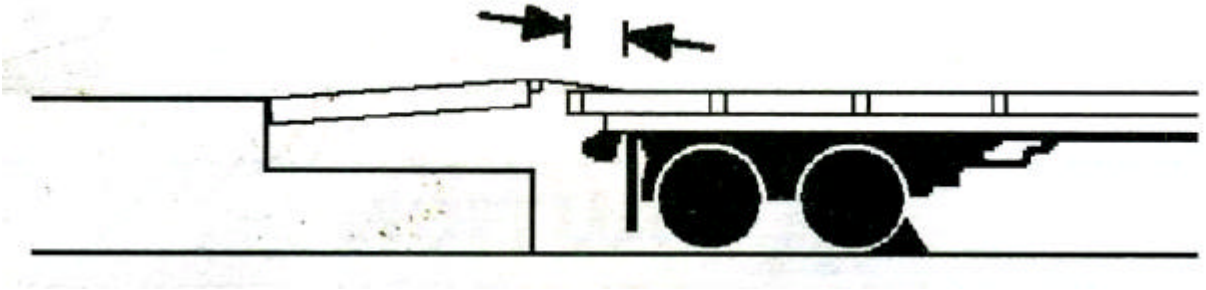
2.4.3 Keep hands and feet clear of pinch pints. Avoid putting any part of your body near moving parts.



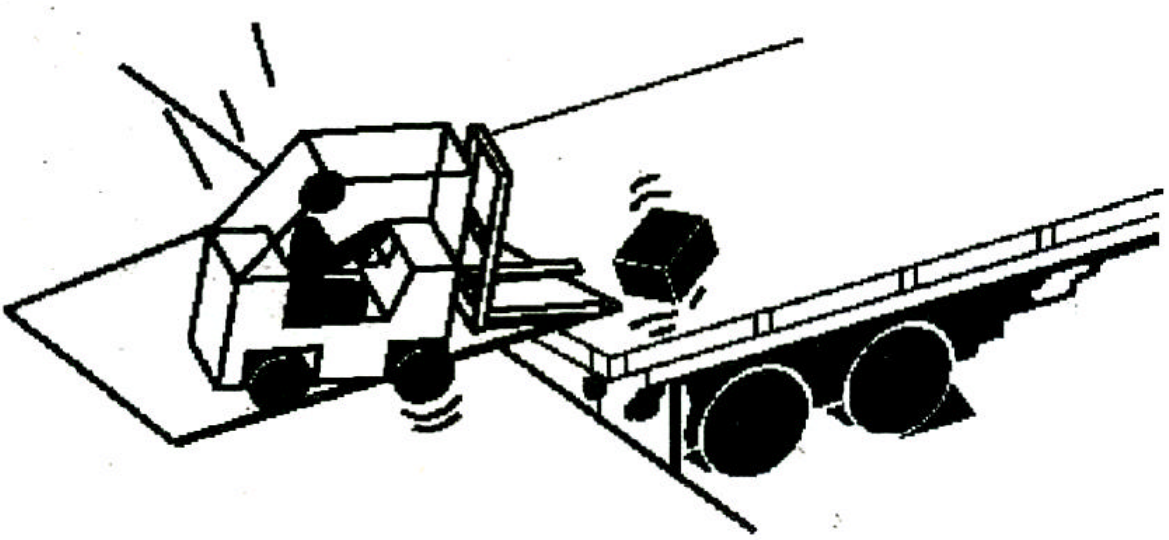
2.4.4 Chock/restrain all freight carriers. Never remove the wheel chocks until loading or unloading is finished and truck driver has been given permission to drive away.



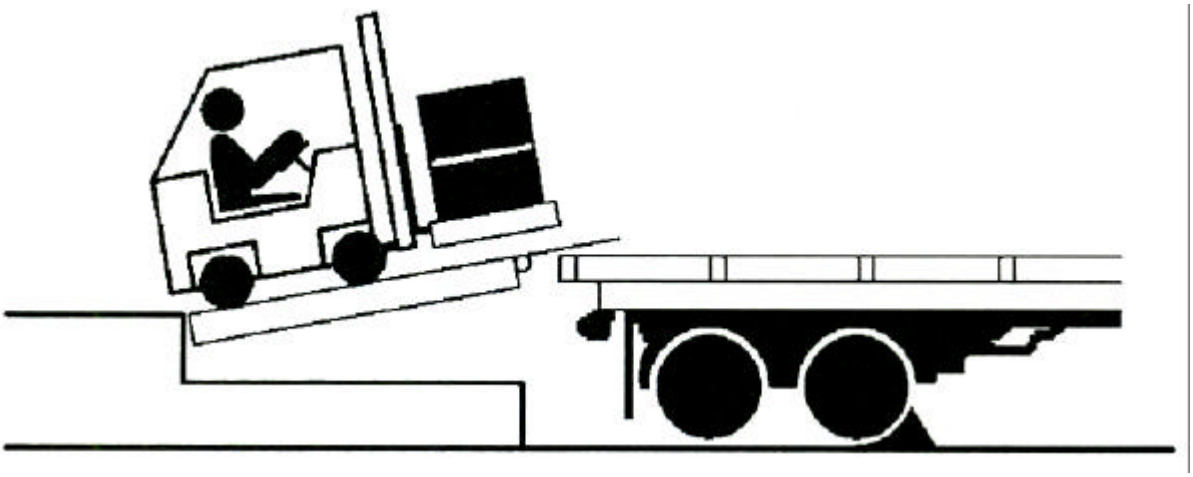
2.4.5 Do not use a broken or damaged dock leveller. Make sure proper service and maintenance procedures have been performed before using.



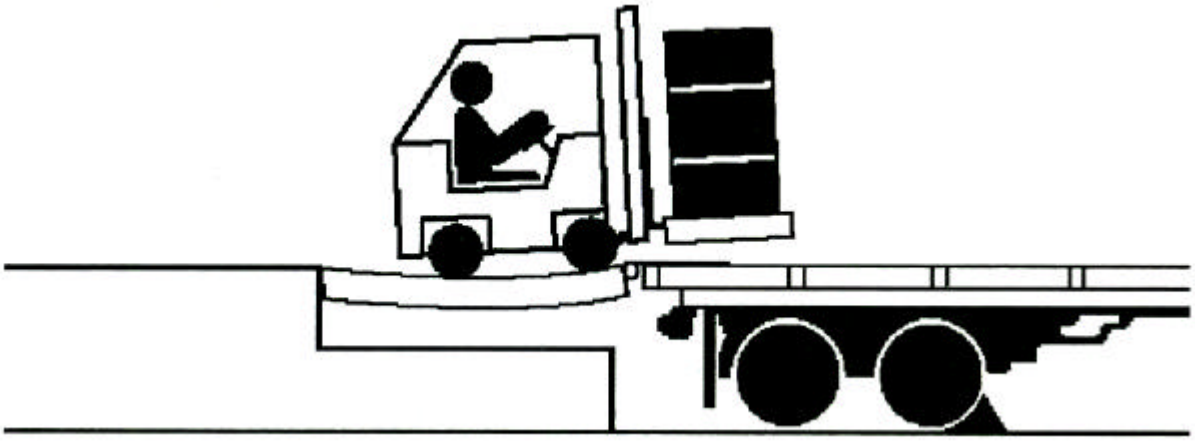
2.4.6 Make sure lip overlaps onto trailer at least 4 in. (100mm)



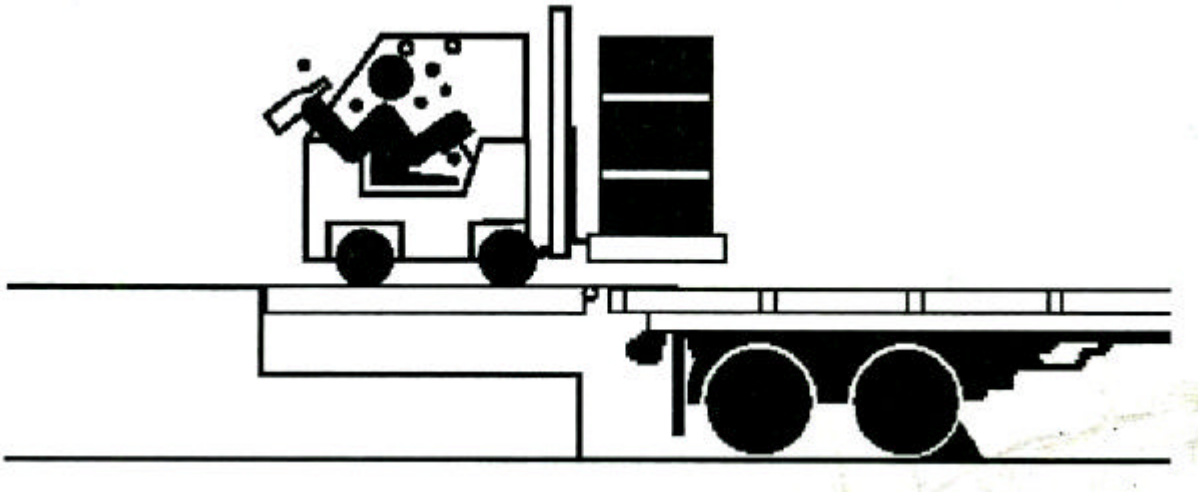
2.4.7 Keep a safe distance from both side edges.



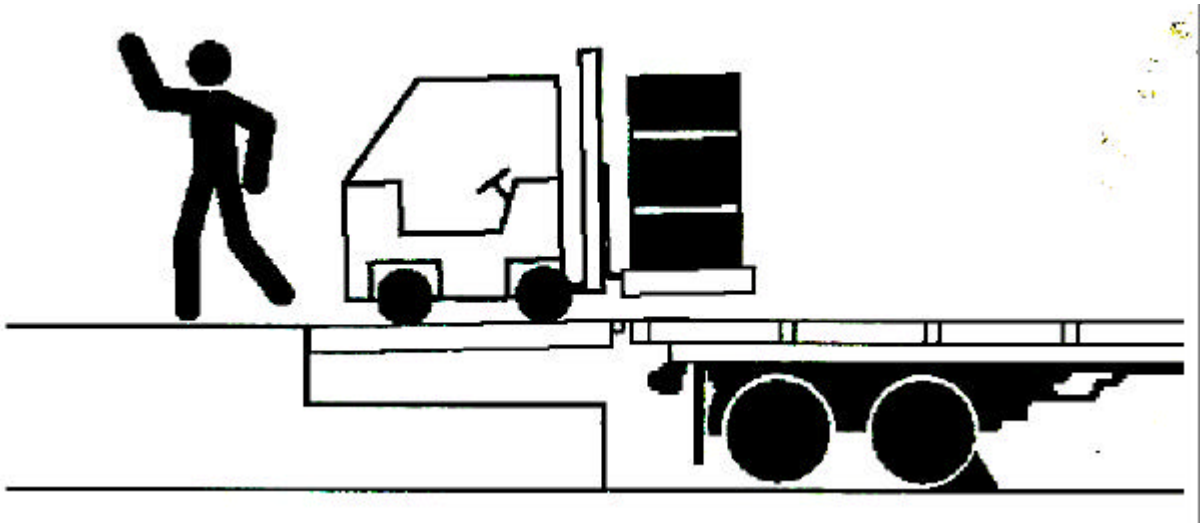
2.4.8 Do not use dock leveller if freight carrier is too high or too low.



2.4.9 Do not overload the dock leveller.



2.4.10 Do not operate any equipment while under the influence of alcohol or drugs.



2.4.11 Do not leave equipment or material unattended on dock leveller.

2.5 Safety symbols

All warning symbols and labels should remain legible at all times. Please note the following:

- ?o clean label use only a damp cloth and a mild soap solution; do not use solvents, petrol, strong detergents etc.;
- ?eplace damaged and/or illegible labels; contact EP for new labels;
- ?eplacement parts must be provided with labels identical to those on original parts.



Safety Decal on Hold-Down Mechanism.

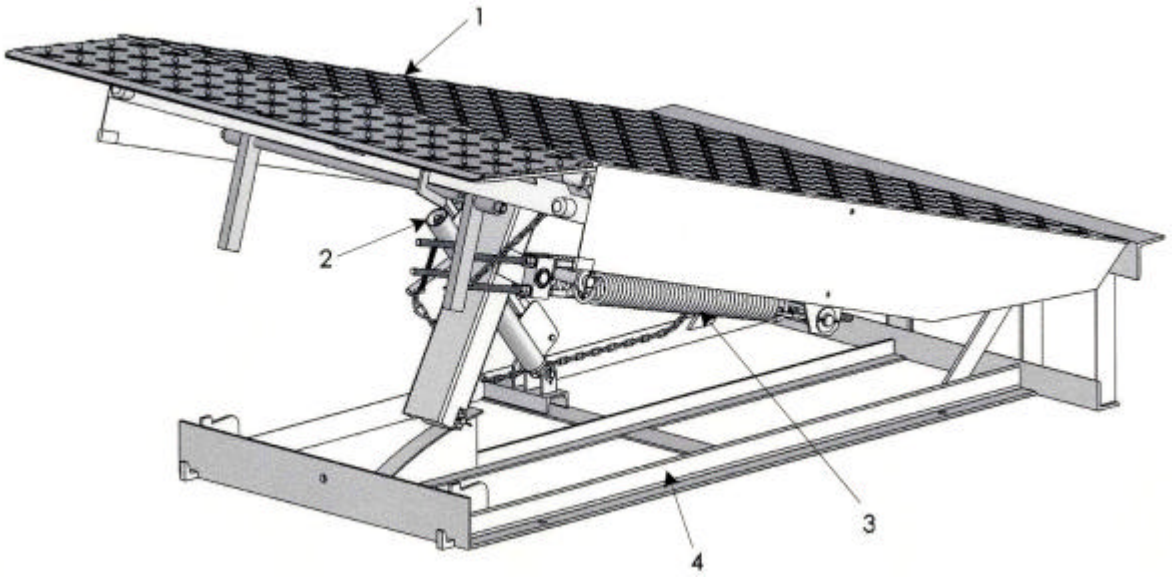
3. GENERAL DESCRIPTION OF THE MACHINE

The Dock Leveler is the special equipment for the cargo loading and unloading, let the operation of the material handling equipments are not influenced by the different height of trailer bed with the platform, such as the forklift,it may drive into the trailer compartment to carry on the handling operation directly, it only requires a single person to complete the process of loading and unloading cargo fast and safely. It can help reduce a lot of the labor cost, improve working efficiency by many times, obtain the better economic efficiency, it is indispensable to the modern warehouse necessary.

EP Mechanical Dock Levelers offer high performance ,minimal maintenance and increased safety for today's busy loading dock. Easy to operate and no heavy lifting required.Simply pull and hold the release chain.The deck is driven up by heavy-duty lift spring and the lip is smoothly powered out to the locked position.The deck is then down onto the trailer bed. When the trailer departs, the lip releases, allowing the unit to be returned to the cross position.

3.1 Components

The Mechanical Dock Leveler (*See* Figures 1) is composed of the Deck mechanism [1],including deck and lip, the Frame [4], the Hold-down mechanism[2] and the Lift mechanism[3].



1.DECK MECHANISM **2.HOLD-DOWN MECHANISM** **3.LIFT MECHANISM** **4. FRAME**

FIG. 1

3.2 Technical features

EP Mechanical dock levellers are available in the following sizes, capacities and options.

Width

6ft (1830mm) / 1800mm

7ft (2110mm) / 2000mm

Legnth

6ft (1830mm) / 2000mm

8ft (2210mm) / 2500mm

10ft(2820mm) / 3000mm

Capacity

25000lbs (12000kg)

30000lbs (14000kg)

35000lbs (16000kg)

Special needs available.

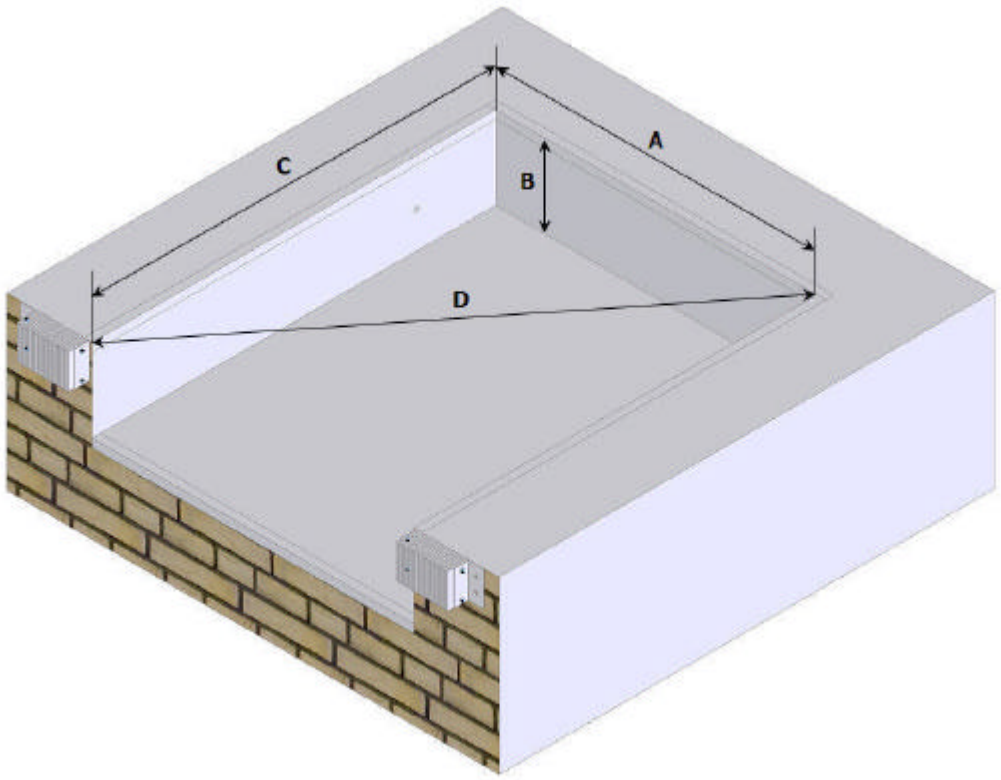
The technical features of Mechanical Dock Levelers listed below. Its main features can be found on the type plate located the front bottom of the Dock Levelers.

1. Model	DKL-12ME-1820 DKL-14ME-1820 DKL-16ME-1820	DKL-25MA-6.8 DKL-30MA-6.8 DKL-35MA-6.8	DKL-12MA-2025 DKL-14MA-2025 DKL-16MA-2025	DKL-25MA-7.8 DKL-30MA-7.8 DKL-35MA-7.8
2. Capacity	12000KG 14000KG 16000KG	25000LB 30000LB 35000LB	12000KG 14000KG 16000KG	25000LB 30000LB 35000LB
3. Platform Size	1800x2000mm		2000x2500mm	
4. Overall Size		72 "x87 "(6 'x 8 ')		83 "x87 "(7 'x 8 ')
5. Height*	600mm	19 "(24 ")	600mm	19 "(24 ")
6. Lip Board Width	450mm	18 "	450mm	18 "
7. Travel (up)	300mm	12 "	300mm	12 "
8. Travel (down)	300mm	12 "	300mm	12 "

*The height means the height in lowered position.

4. INSTALLATION

4.1 Prepare pit



A-Distance (Pit Width) (Front and Rear) **B-Distance (Deck Floor-to Pit Floor) (All Four Corners)** **C-Distance (Pit Length) (Both Sides of Pit)** **D-Distance (Pit Corner-to-Corner) (Top, Bottom, and Both Sides**

Fig. 2

Dimensions for Pit

Leveler Size	Dim "A" width	Dim "C" length	Dim "B" depth (rear)	Dim "B" depth (front)
6 'x 6 '	74 '	63 '	19 1/2 '	20 '
7 'x 6 '	85 '	63 '	19 1/2 '	20 '
6 'x 8 '	74 '	87 '	19 1/2 '	20 '
7 'x 8 '	85 '	87 '	19 1/2 '	20 '
6 'x 10 '	74 '	111 '	19 1/2 '	20 '
7 'x 10 '	85 '	111 '	19 1/2 '	20 '

Leveler Size (width x length)	Dim "A" width	Dim "C" length	Dim "B" depth (rear)	Dim "B" depth (front)
1800mm x 2000mm	1860mm	2150mm	600mm	605mm
2000mm x 2000mm	2060mm	2150mm	600mm	605mm
2000mm x 2500mm	2060mm	2650mm	600mm	605mm
2000mm x 3000mm	2060mm	3150mm	600mm	605mm

Warning
Post safety warning and barricade the work area at dock level and ground level to prevent unauthorized use of the dock leveller before installation has been completed.

Warning
Only trained installation professionals with the proper equipment should install this product.

Warning
Do not remove the shipping bolts fixed on the dock leveller lip until instructed to do so.

Before lowering the dock leveller into the pit, the following must be performed,

1. Remove all debris from the pit and sweep the pit clean
2. Check the entire dock leveller pit for proper construction according to approved / certified pit drawing. Make sure pit is square by making the following measurements,

- Measure pit width distance A at both front and rear of pit.
- Measure dock floor-to-pit floor distance B at all four corners
- Measure pit length distance C at both sides
- Measure corner-to-corner(cross-cross) distance D at both sides. Take measurements at dock floor level and at pit floor level.

4.2 Installation

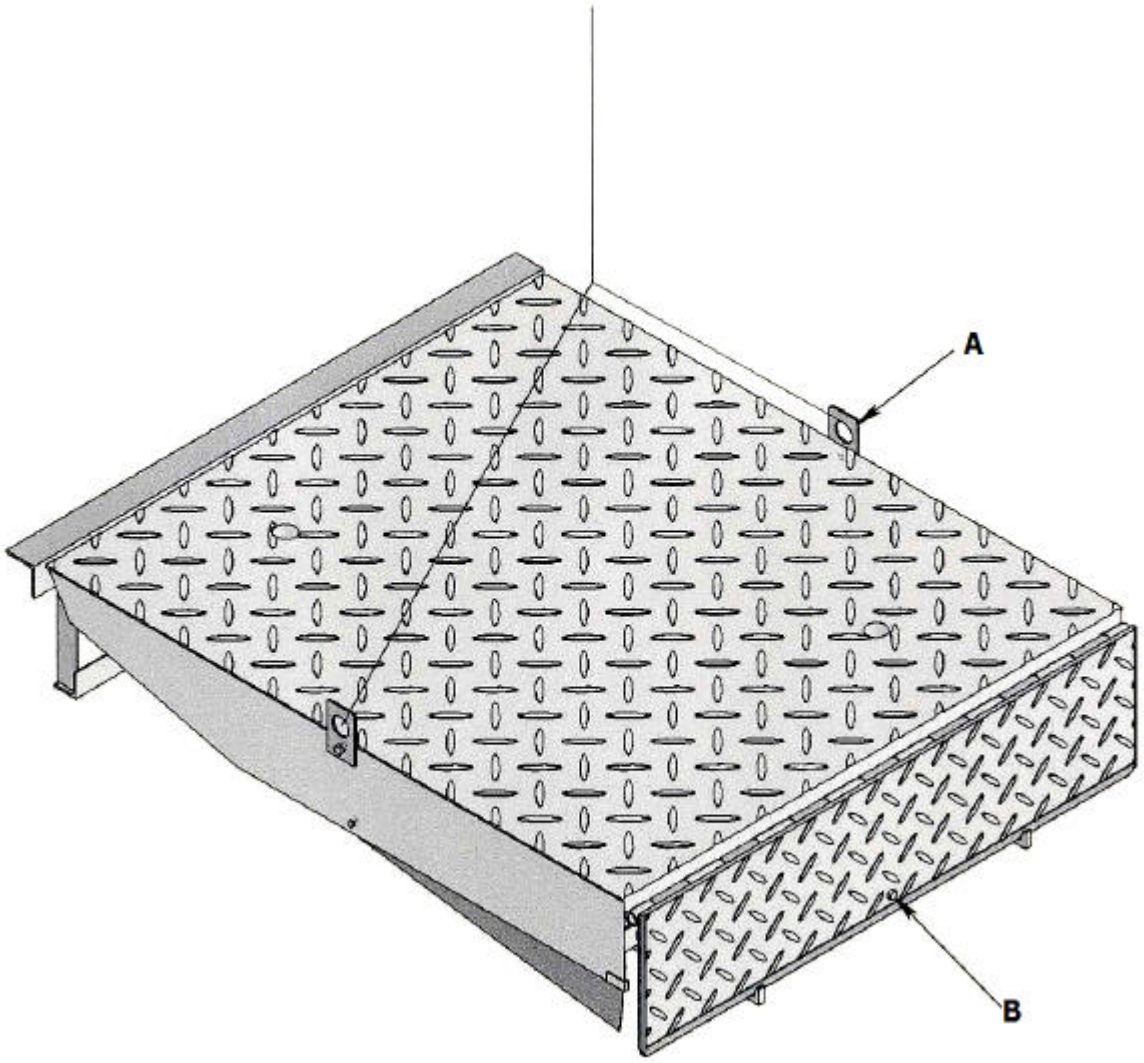
EP dock levellers are designed with installation in mind. Each unit is shipped with lifting bracket A fastened to the platform side joists.

Using a lifting device and chains with the appropriate lifting capacity and reach, Always use the lifting brackets whenever lowering or lifting a dock leveller into or out of a pit.

Before lifting, remove any bumpers that may be banded or bolted to the frame of the dock leveller.

Warning

Do not remove the shipping bolts B fixed on the Dock Leveller lip and frame this time.The shipping bolts are needed to hold the leveller together during the installation process.



A-LIFTING BRACKET B-SHIPPING BOLT

FIG. 3

When the machine is placed in a pit floor and firmly anchored to it with 4 expansion bolts. Use the data in the table below as a guide:

Bolt		**Expanding tube**		**Hole**	
Spec.	*Length*	*Outer diameter*	*Length*	*Diameter*	*Depth*
M16	150 mm	22 mm	90 mm	23 mm	100 mm

5. OPERATING INSTRUCTIONS

⚠ **warning**

- Stay clear of dock leveller when freight carrier is entering or leaving dock area.
- Do not raise or lower the dock leveller if anyone is under or in front of leveller.
- Keep hand and feet clear of pinch pints. Avoid putting any parts of your body near moving parts.
- Do not active the dock leveller until the truck/trailer is positioned squarely against the bumper and the truck wheels are chocked.
- Do not use the dock leveller if the freight carrier is too high or too low.(as **FIG. 4)**
- Never remove the wheel chocks until loading an unloading is finished and truck driver has been given the permission to leave.

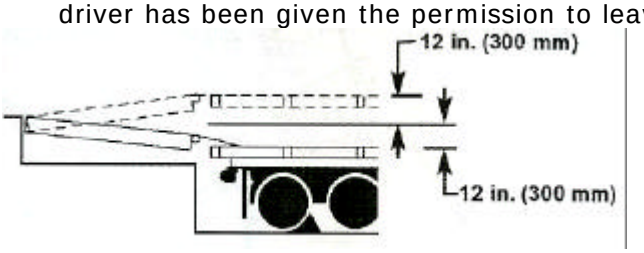


FIG. 4

The work range of EP dock leveller is ±12 inch (300mm).

The instructions of loading/unloading

1. Make sure truck/trailer is positioned squarely against dock bumpers.
2. Instruct driver to remain at the dock until the loading or unloading process has been completed.
3. Chock the truck/trailer wheels or use truck restraint if present.
4. Raise the platform by pulling and holding the platform release chain (A).(as **FIG. 5**).

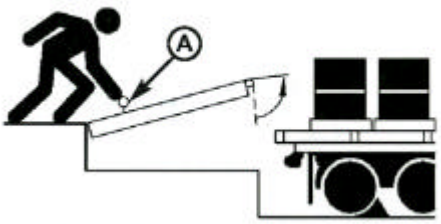


FIG. 5

5. Hold the release chain until the platform is at the fully-raised position and lip is fully extended.
6. Walk out onto the platform. The platform will lower until the lip rests on the truck/trailer bed.
   (if the truck/trailer bed is below dock level, pull and hold the pull chain (B). as **FIG. 6**)

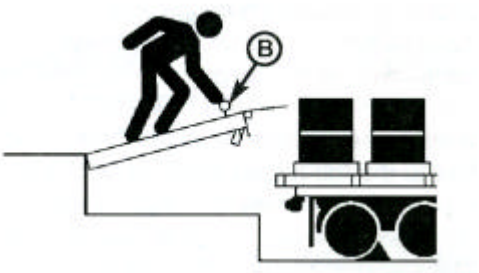


FIG. 6

7. Make sure that the lip is fully extended and supported on the truck/trailer along the entire width of the platform with at least 4 inch (100mm) of lip contacting the truck bed.
8. Proceed with loading or unloading.



FIG. 7

9. When the loading or unloading process has been finished, return platform to cross-traffic (stored) position as follow,

- Raise the platform to the full-raised position by pulling and holding the platform chain(A).
- Remove chocks from truck/trailer wheels or release truck restraint.
- Indicate to driver that truck should leave the dock.
- Lower platform by slowly walking out onto the platform until platform lowers to the cross-traffic position. Then put up the lip little, loosen quickly, let the lip engages in the lip keepers.

6. TRANSPORT

- With forklift truck: drive the forks of the forklift truck into the fork slots at the rear of the Dock Levelers
- With crane: lift the dock leveler by lifting bracket

As shown on the **FIG. 8**

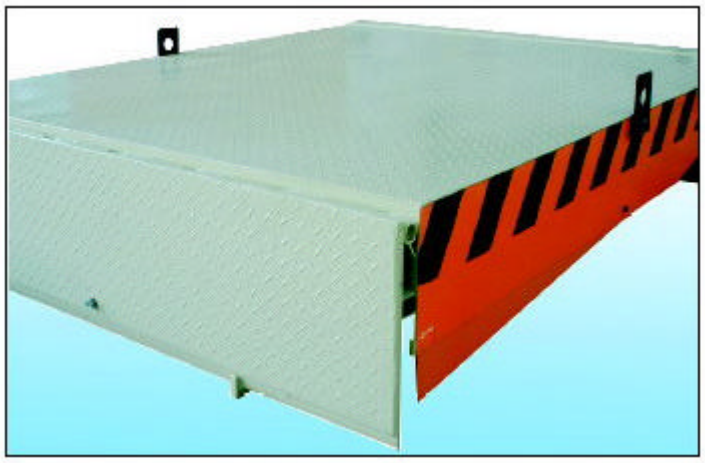


FIG. 8

note

When use forklift truck to transport the Dock Leveler, please notice the fork slots is at rear position, DO NOT drive fork from the sides of Dock Leveler.

7. STORAGE

Cover components that are sensitive to moisture. Apply an anti-corrosive agent to parts that rust easily.

8. MAINTENANCE INSTRUCTIONS

8.1 Important advice

In order to keep your Dock Levelers in a safe and optimum condition for as long as possible, it is important to make sure repair reports are carefully updated. Always keep a daily maintenance book with the Dock Leveler (*see* Annex I). This must be updated at the intervals determined by the operator's maintenance procedures, i.e. daily, weekly etc.

warning

- Before starting maintenance or repair work: read the safety measures in Chapter 4 and cut the power.
- Your Dock Levelers must only be maintained by trained and authorized personnel.
- Check the Dock Levelers daily before start up.Never use a Dock Leveler with a fault.
- Always post safety warning and barricade the work area at dock level and ground level to prevent unauthorized use of the dock leveller before maintenance is complete. Failure to do this may result in serious personal injury or death.
- Always stand clear of the dock leveller lip when working in front of the dock leveller. Failure to do this may result in serious personal injury or death.
- The platform maintenance prop must be in the serve position when working under the dock leveller. (as **FIG.9**)

Note

- Repair faults immediately. Only use original spare parts or parts of equivalent quality.
- All spent lubricants and waste products must be disposed of in line with local environmental laws and controls (e.g. waste oil, used batteries, worn parts etc.).
- It may be useful to consult the "Troubleshooting Guidelines" in Annex II.

Contact EP Service & Repair Division if you have any technical questions or if your Dock Leveler has a complex fault.

Identify your Dock Leveler by quoting the EP number on the type plate so that we can help you as quickly as possible.

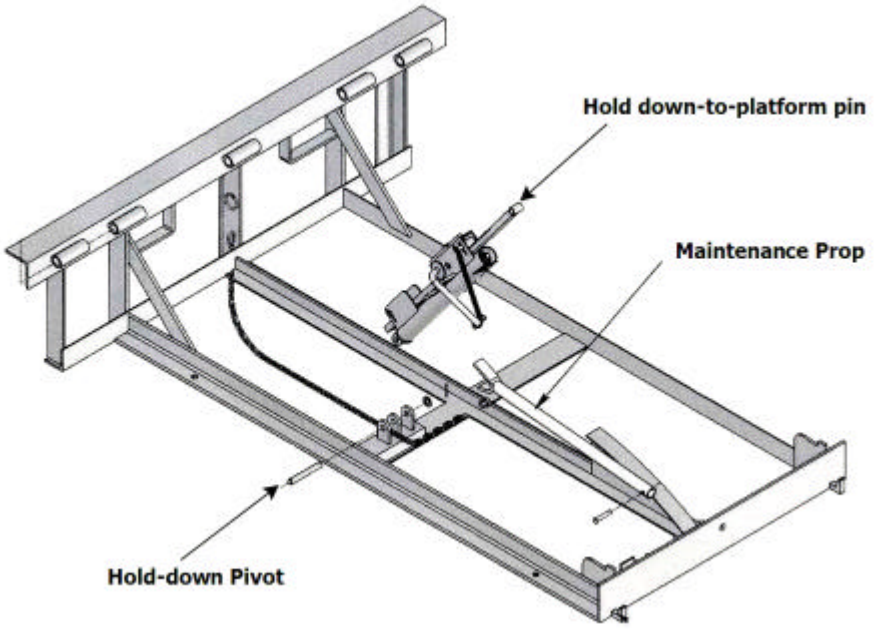


FIG. 9

8.2 Maintenance and Inspections

Weekly Maintenance

- Operate the dock leveller through the complete operating cycle to maintain lubrication.
- Inspect the platform hinge and the lip hinge areas. The hinge areas must be kept free of dirt and debris. Build-up of foreign material in the hinge areas will increase wear and cause abnormal operation.

⚠ **warning**

Failure to properly lubricate the dock leveller will cause abnormal operation of the leveller.
Do not lubricate the hold-down ratchet bar and pawl.

Quarterly Maintenance

- Lubricate the following areas with light-weight machine oil,
  -Hold-down Pivot
  -Hold down-to platform pin
  (as **FIG.9**)

- Lubricate the following areas with white lithium grease
  -Lip hinge area (inject grease into all the lip hinge grease fittings) (as **FIG.10**)
  -Platform hinge area (inject grease into all the platform hinge)
  -Lift arm roller bushing
  -Lift arm pivot
  -Lip assist pin



FIG.10

ANNEX I DAILY CHECKLIST

Report and repair defects before putting the dock levelers into service

OPERATOR:
DOCK LEVELERS ID:
DATE:

Checkpoints	OK	not OK	Remarks
1. Mechanical system			
Springs			
Ratchet bar			
Chains			

Operator remarks:

EP SERVICE & REPAIR DIVISION
Hangzhou,China

ANNEX II TROUBLESHOOTING (Mechanical Dock Levelers)

FAULT	POSSIBLE CAUSE	SOLUTION
Platform does not rise	Heavy object on platform.	Remove object from platform.
	Disconnected or broken release chain.	Connect or replace chain.
	Shipping bolts fixed	Remove the shipping bolts.
Platform rises slowly or does not rise to the full raised position.	Heavy object on platform.	Remove object from platform.
	Insufficient main spring tension.	Increase tension on main spring.
	Damaged or worn hold-down mechanism.	Clean and inspect hold-down mechanism. Or replace the hold-down mechanism.
Platform rises to full height, but lip does not fully extend.	Insufficient lip spring tension	Increase tension on lip spring.
	Disconnected lip assist bar.	Connect lip assist bar.